Unified MEGA Mathematical Framework

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Abstract

This document unifies the original MEGA mathematical model with the behavior of the current reference implementation. It includes the complete hash encoding system, geometric treatment (hyperboloid model), operator semantics exactly matching the code: open inside emits End ... Start, delimiter-aware crossover forbids cuts inside protected regions, mutation weights are spelled out, initialization is documented, and identifier accounting is piecewise-constant with reuse. Terminology is standardized: delimited regions are inside matching [Start ... End]; undelimited positions are outside all delimiter pairs.

1 Hash Encoding System

1.1 Core Hash Function

Let $H : \mathbb{N} \to \mathbb{Z}_{64}$ be the hash function defined by:

$$H(i) = \text{xxhash.xxh64_intdigest}(\text{str}(i))$$

This provides a deterministic mapping from integer identifiers to 64-bit hash keys.

1.2 Reserved Delimiter Encodings

The system maintains fixed delimiter assignments:

$$\varepsilon_{\text{start}} = H(1) \quad \text{(Start delimiter)} \tag{1}$$

$$\varepsilon_{\text{end}} = H(2) \quad \text{(End delimiter)} \tag{2}$$

1.3 Identifier Management

Let $c(t)$ denote the gene counter at time t, initialized as $c(0) = 3$. The system maintains:

- **Active encodings**: $\mathcal{E}(t) = \{H(i) \mid i \text{ is assigned at time } t\}$
- **Available pool**: $\mathcal{P}(t) = \{\varepsilon \mid \varepsilon \text{ freed and available for reuse}\}$
- **Counter invariant**: $|\mathcal{E}(t)| + |\mathcal{P}(t)| \leq c(t) - 3$

1.4 Gene Encoding Process

For a new gene g:

Algorithm 1 Gene Encoding Assignment

```
if g ∈ reverse_encodings then
    return existing hash key
else if |P(t)| > 0 then
    ε ← P(t).pop()
    return ε
else
    id ← c(t)
    c(t + 1) ← c(t) + 1
    return H(id)
end if
```

2 Base Search Space

Let $G = \{g_1, g_2, \ldots, g_k\}$ be a finite set of base gene strings, where each g_i maps to hash key $\varepsilon_i = H(\mathrm{id}_i)$ for $\mathrm{id}_i \geq 3$.

Define the extended alphabet:

$$\tilde{G} = G \cup \{\mathrm{Start}, \mathrm{End}\} \cup \{0\}$$

where 0 represents the absence of a gene (for dimensional consistency in variable-length solutions).

Let $n \geq 1$ be the maximum solution length. Define the primary combinatorial search space:

$$V = \tilde{G}^n, \quad |V| = (k+3)^n$$

Each $x \in V$ is a sequence $(x_1, \ldots, x_n)$ with $x_i \in \tilde{G}$.

3 Parameter Space

Let $\Pi = \{\pi_1, \ldots, \pi_{11}\}$ denote system parameters:

3.1 Population Parameters

$$\pi_1 : \text{population size} \tag{3}$$
$$\pi_2 : \text{number of parents} \tag{4}$$
$$\pi_3 : \text{maximum individual length} \tag{5}$$

3.2 Mutation Parameters

$$\pi_4 : \text{mutation probability} \tag{6}$$
$$\pi_5 : \text{delimited-region mutation probability} \tag{7}$$
$$\pi_6 : \text{delimiter-delete probability} \tag{8}$$
$$\pi_7 : \text{open probability} \tag{9}$$
$$\pi_8 : \text{metagene (capture) probability} \tag{10}$$
$$\pi_9 : \text{delimiter-insert probability} \tag{11}$$

3.3 Gene Selection Parameters

$$\pi_{10} : \text{base-gene probability} \tag{12}$$

$$\pi_{11} : \text{metagene probability (exponential weighting factor)} \tag{13}$$

3.4 Initialization Process

Each initial organism length $L \sim \text{Uniform}\{2, \ldots, \pi_3\}$. Genes are i.i.d. from the base pool. If delimiter injection is enabled, insert [Start, End] pairs with probability 0.10 at candidate sites, obeying a spacing constraint.

4 Metagenes and Compressed Representations

4.1 Metagene Definition

Let $\mathcal{M} = \{M_1, \ldots, M_m\}$ be the set of metagenes. Each M_i is represented as:

$$M_i = (\varepsilon_{i1}, \varepsilon_{i2}, \ldots, \varepsilon_{i\ell_i})$$

where $\ell_i = |M_i|$ and each ε_{ij} is a hash key.

4.2 Metagene Encoding

For metagene M_j, assign identifier id_j and encoding $\varepsilon_j = H(\text{id}_j)$ such that:

$$\text{encodings}[\varepsilon_j] = (\varepsilon_{j1}, \varepsilon_{j2}, \ldots, \varepsilon_{j\ell_j})$$

4.3 Expansion Function

Define expansion function $\xi : \mathcal{M} \to G^*$ where $\xi(M_i)$ recursively expands all metagene references within M_i to base gene sequences.

4.4 Compressed Search Space

Extend the alphabet: $\tilde{G} = G \cup \mathcal{M} \cup \{\text{Start}, \text{End}\} \cup \{0\}$.

For each $x \in V$, define $R(x)$ as all representations obtained by zero or more metagene compression operations ϕ_{M_i} that replace contiguous subsequences matching $\xi(M_i)$ with the single symbol M_i.

5 Network Structure and State Space Dynamics

5.1 Network Definition

Let $\mathcal{G}(t) = (\mathcal{M}(t), \mathcal{E}_g(t))$ be the metagene dependency graph where:

- $\mathcal{M}(t)$: active metagenes at time t
- $\mathcal{E}_g(t) = \{(M_i, M_j) \mid M_i \text{ appears in } \xi(M_j)\}$: dependency edges

Define:

- $d(M_i)$: maximum path length from M_i to any base gene
- $u(M_i, t)$: number of active references to M_i at time t
- $D(M_i, t) = \{M_j \mid M_i \in \xi(M_j) \wedge u(M_j, t) > 0\}$: dependents of M_i

5.2 Enhanced State Spaces

Each encoding identifier $e \in \mathcal{E}$ resides in exactly one of three spaces:

- $\mathcal{A}(t)$: active (LRU tracked)
- $\mathcal{T}(t)$: deletion consideration basket
- $\mathcal{P}(t)$: available pool for reuse

State conservation: $\forall t, \forall e \in \mathcal{E} : e \in \mathcal{A}(t) \cup \mathcal{T}(t) \cup \mathcal{P}(t)$ and these sets are pairwise disjoint.

5.3 State Transition Operators

- **LRU eviction**: $\phi_T : \mathcal{A}(t) \to \mathcal{T}(t)$ moves least-used items subject to dependency constraints
- **Deletion with preservation**: $\phi_P : \mathcal{T}(t) \to \mathcal{P}(t)$ after $\theta = 2$ generations inlines $\xi(M_i)$ into dependents then frees the identifier
- **Reactivation**: $\phi_A : \{\mathcal{T}(t) \cup \mathcal{P}(t)\} \to \mathcal{A}(t)$ reintroduces an identifier

6 Geometric Interpretation and Search Space Warping

6.1 Hyperbolic Model (Hyperboloid)

Let $\mathbb{H}^n = \{x \in \mathbb{R}^{n+1} : \langle x, x \rangle_M = -1, x_0 > 0\}$ be the hyperboloid model with Minkowski inner product:

$$\langle x, y \rangle_M = -x_0 y_0 + \sum_{j=1}^{n} x_j y_j$$

Define embedding $\phi : V \to \mathbb{H}^n$ by:

$$\phi(x) = \left(\sqrt{1 + \|x\|^2}, x_1, \ldots, x_n\right)$$

and return map $\psi : \mathbb{H}^n \to \mathbb{R}^n$:

$$\psi(x_0, \ldots, x_n) = (x_1, \ldots, x_n)$$

6.2 Distance Metrics

- **Euclidean distance**: $d_E(x, y) = \|x - y\|$
- **Hyperbolic distance**: $d_H(x', y') = \operatorname{arccosh}(-\langle x', y' \rangle_M)$ for $x', y' \in \mathbb{H}^n$

6.3 Metagene Transformations

Each M_i induces an isometry $T_i = \exp(B_i)$ on $\mathbb{H}^n$. For composite transformation $T = T_m \circ \cdots \circ T_1$, define transformed Euclidean distance:

$$d'_E(x, y) = \|\psi(T(\phi(x))) - \psi(T(\phi(y)))\|$$

6.4 Volume and Compression

Hyperbolic volume ratio:

$$V_H(P_t) = \frac{\mathrm{Vol}(T(\phi(P_t)))}{\mathrm{Vol}(\phi(P_t))}$$

Enhanced compression ratio:

$$CR'(P_t) = \frac{V'_{\text{eff}}(P_t)}{|V|}$$

where $V'_{\text{eff}}(P_t) = |G|^{\mathrm{Dim}'_{\text{eff}}(P_t)}$.

7 Network Decay and Stability

For a connected component $C \subseteq \mathcal{G}(t)$, define inactivity measure:

$$I(C, t) = \prod_{M_i \in C} (1 - \min(1, u(M_i, t)))$$

Then the probability of deletion of component C at time $t + \theta$ is:

$$P(\text{deletion of } C \text{ at } t + \theta) = I(C, t)^\theta$$

Information is preserved if $\exists M_j \in D(M_i, t) : u(M_j, t) > 0$.

8 Temporal Evolution and Dynamic Dimensionality

Let τ be the time unit, $t \in \mathbb{N}$. Define:

- $s(t) = \lfloor t/\text{max_length} \rfloor \bmod \pi_1$: schedule function
- $p(t) = t \bmod \text{length}(s(t))$: position function
- $n(t) = \max\{\text{len}(x) \mid x \in P_k, k \leq t\}$: dynamic dimension

Then $V(t) = G^{n(t)}$ with $|V(t)| = (k + 3)^{n(t)}$.
Each time step τ, increase $u(M_i, t)$ by τ for all $M_i \in \mathcal{A}(t)$.

9 Mutation Framework

9.1 Context Function

Define context function $C : \mathbb{N} \times V \to \{0, 1, 2\}$:

$$C(i, x) = \begin{cases} 0 & \text{if position } i \text{ is undelimited} \\ 1 & \text{if position } i \text{ is delimited} \\ 2 & \text{if } x_i \in \{\text{Start}, \text{End}\} \end{cases} \tag{14}$$

9.2 Operator Sets

Define context-dependent operator sets:

$$\mathcal{M}_0 = \{\text{point, swap, insertion, deletion, insert_delimiter_pair, open}\} \tag{15}$$
$$\mathcal{M}_1 = \{\text{point, swap, insertion, deletion, capture, open_no_delimit}\} \tag{16}$$
$$\mathcal{M}_2 = \{\text{delimit_delete, swap}\} \tag{17}$$

9.3 Mutation Weights

Assign probabilities via weight function $W_c : \mathcal{M}_c \to \mathbb{R}_+$:

For undelimited context ($C(i, x) = 0$):

$$W_0 = (1, 1, 1, 1, \pi_9, \pi_7)$$

over $\mathcal{M}_0$.

For delimited context ($C(i, x) = 1$):

$$W_1 = (1, 1, 1, 1, \pi_8, \pi_7)$$

over $\mathcal{M}_1$.

For delimiter tokens ($C(i, x) = 2$):

$$W_2 = (\pi_6, 1 - \pi_6)$$

over $\mathcal{M}_2$.

9.4 Gene Selection with Exponential Weighting

For metagene selection from stack of size m, assign weights:

$$w_i = \pi_{11}^{m-i-1}, \quad i = 0, 1, \ldots, m-1$$

Normalized probabilities:

$$p_i = \frac{w_i}{\sum_{j=0}^{m-1} w_j}$$

10 Modified Open Mutation Operator

10.1 Context-Dependent Open Semantics

For a metagene M at position i with context $C(i, x)$:

Outside delimiters ($C(i, x) = 0$):

$$\text{open}(M) = [\text{Start}] \oplus \xi(M) \oplus [\text{End}]$$

Inside delimited region ($C(i, x) = 1$):

$$\text{open_no_delimit}(M) = [\text{End}] \oplus \xi(M) \oplus [\text{Start}]$$

10.2 Delimited Region Transformation

For a delimited region D containing metagene M:

$$D = [\text{Start}, g_1, \ldots, g_k, M, h_1, \ldots, h_m, \text{End}]$$

The open operator transforms D into:

$$D_1 \oplus \xi(M) \oplus D_2$$

where:

$$D_1 = [\text{Start}, g_1, \ldots, g_k, \text{End}] \tag{18}$$
$$D_2 = [\text{Start}, h_1, \ldots, h_m, \text{End}] \tag{19}$$

10.3 Properties

1. **Conservation**: $|D_1| + |\xi(M)| + |D_2| = |D| + 4$
2. **Delimiter integrity**: $\gamma(D) = 1 \Rightarrow \gamma(D_1 \oplus \xi(M) \oplus D_2) = 2$
3. **Pattern accessibility**: each $g \in \xi(M)$ becomes undelimited

11 Evolution Operators

11.1 Crossover (Delimiter-Aware)

Define $\text{Ind}_U(x_1, x_2)$ as the set of indices outside all [Start ... End] pairs in both parents x_1, x_2.

Crossover point selection:

$$\text{crossover_point} = \begin{cases} \text{Uniform}(\text{Ind}_U(x_1, x_2)) & \text{if } |\text{Ind}_U(x_1, x_2)| > 0 \\ \text{None} & \text{otherwise} \end{cases}$$

11.2 State Transition with Open

Let Φ_τ denote the one-step evolution operator. With probability $P(\text{open} \mid p(t))$, apply ϕ_{open}; with probability $P(\mu \mid p(t))$, apply mutation; otherwise preserve $x_{p(t)}$.

11.3 Search-Space Impact

For sequence $s = s_1 \oplus D \oplus s_2$, the open transformation yields:

$$s_1 \oplus D_1 \oplus \xi(M) \oplus D_2 \oplus s_2$$

increasing local dimensionality, creating new capture opportunities at D_1, D_2 boundaries, and maintaining global nesting properties.

12 Conservation Laws and Invariants

12.1 Information Conservation

Total information is preserved by reversible metagene expansions:

$$I(x) = I(R(x))$$

for any representation $R(x)$ of sequence x.

12.2 Structural Integrity

Define balance function $\beta(x) = \sum_i (\mathbf{1}_{\text{Start}}(x_i) - \mathbf{1}_{\text{End}}(x_i))$.

Valid sequences satisfy:

1. $\beta(x) = 0$ (balanced delimiters)
2. $1 \leq \text{len}(x) \leq \pi_3$ (length constraints)
3. All prefixes have $\beta(\text{prefix}) \geq 0$ (proper nesting)

12.3 Volume Conservation

Within fixed dimension: $I(x) = I(R(x))$. Expansion arises via dimension growth or metagene deletion.

12.4 Identifier Accounting

$$|\mathcal{A}(t)| + |\mathcal{T}(t)| + |\mathcal{P}(t)| = |\mathcal{E}(t)|$$

except that fresh capture when $\mathcal{P}(t) = \emptyset$ increments total by 1; otherwise reuse maintains constant count.

12.5 Network Conservation

$\mathcal{A}(t)$ forms a connected subgraph; hierarchy is maintained through dependency preservation.

13 Overlapping and Nested Metagenes

For overlapping/nested metagenes, consider submanifolds:

$$S_i = \{x' \in \mathbb{H}^n \mid x' = \phi(x), x \in V_{M_i}\}$$

Define interaction measure:

$$I(M_i, M_j) = \frac{\text{Vol}(S_i \cap S_j)}{\min(\text{Vol}(S_i), \text{Vol}(S_j))}$$

14 Evolution of Knowledge

Define effective dimension:

$$\dim_{\text{eff}}^{(t)}(x) = \min_{T \in \mathcal{T}^{(t)}(x)} \left[n(t) - \sum_{M_i \in \mathcal{A}(t)} c_i(T)(\ell_i - 1) \right]$$

where $\mathcal{T}^{(t)}(x)$ denotes valid transformations at time t.

The LRU cache preserves frequent patterns; $\mathcal{M}(t+1) = \mathcal{M}(t) \cup \text{New}_t$ or removes D_t. Typically $\dim_{\text{eff}}^{(t)}(x)$ decreases as patterns emerge.

15 Summary

MEGA unifies evolutionary search with dynamic pattern formation through:

- Base space $V = \tilde{G}^n$ with hash encoding $H : \mathbb{N} \to \mathbb{Z}_{64}$
- Emergent metagenes $\mathcal{M}$ with dependency graph $\mathcal{G}(t)$
- LRU state triple $(\mathcal{A}, \mathcal{T}, \mathcal{P})$ with $\theta = 2$ deletion threshold
- Hyperbolic geometry providing natural lens for hierarchy analysis
- Context-dependent mutation operators with delimiter awareness
- Information-preserving evolution maintaining structural invariants

The framework bridges implementation mechanics (hash encoding, LRU cache, operator weights) with analytical tools (hyperbolic geometry, network theory, compression metrics) to provide both algorithmic specification and theoretical understanding of emergent pattern dynamics.